SPECIAL SESSION · 7 SLIDES · BANKING & FINANCE

Graph Engineering

Prompt, context, harness and loop are four disciplines, and each fails in its own way. When the work has an order you need to draw, checkpoint and pause, a fifth one takes over.

Graph Engineering | Not Graph RAG | LangGraph 1.0 | StateGraph | Checkpoints

KGP Talkie
LEARN. BUILD. TALK AI

Scan to subscribe

Five Jobs, All Called Prompt Engineering

THE DISCIPLINE	WHAT IT LOOKS LIKE WHEN THIS IS THE BROKEN ONE	SCOPE
Prompt What do I say?	The model ignores a rule that is sitting right there in the message, or obeys it four times in five.	● ○ ○ ○ ○
Context What goes in the window, what stays out?	A fluent, confident answer about the wrong quarter. Nothing is malformed. The evidence was simply never there.	● ● ○ ○ ○
Harness What does the code around the model do?	Unparseable JSON at 3 AM, a timeout with no retry, a tool call nobody validated, a fallback nobody wrote.	● ● ● ○ ○
Loop How many calls, and what says the work is done?	It never stops, or it stops one step too early. The invoice is usually the first place anyone notices.	● ● ● ● ○
Graph How do several loops coordinate?	Node two wrote something uninformed and node five treated it as fact. By the time it shows, it has flowed through half the system.	● ● ● ● ●

When One Loop Is No Longer Enough

LOOKS LIKE a workflow engine. LangGraph shipped exactly this in January 2024.

The name is new. The practice is not.

ONE LOOP

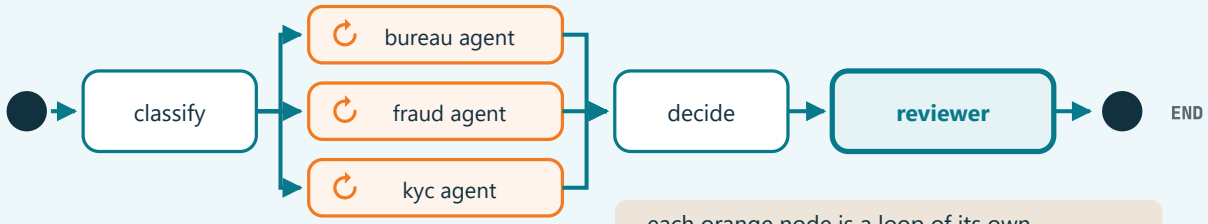
one job, one finish line, one thing checking it



a one-node graph, with an edge back to itself

SEVERAL LOOPS, COORDINATED

what runs when, in what order, and who checks whom



each orange node is a loop of its own

REACH FOR A GRAPH WHEN

GENUINE SPECIALTIES

a different model, a different toolset, or a genuinely separate role

FAN-OUT AND JOIN

independent work that should run at once, then be merged

A MODEL PER STEP

a cheap one to classify, a strong one to draft, a third one to review

FAILURE ISOLATION

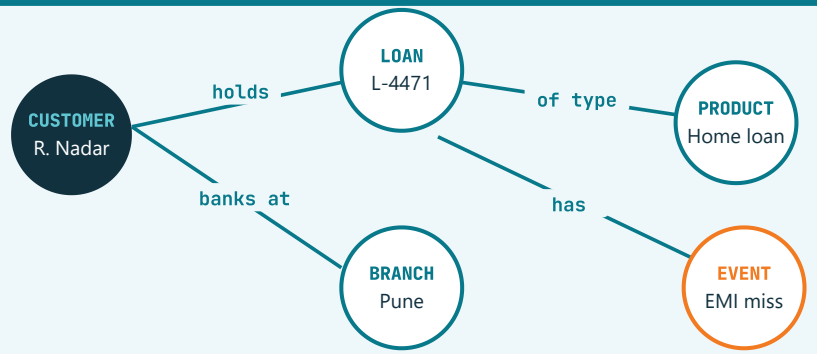
one step can fail without poisoning the rest, and the route is auditable

Otherwise stay in the loop. LangGraph's own guidance says that if your agent is a straightforward loop with tools, LangGraph is overkill.

This Is Not Graph RAG

GRAPH RAG

a graph of your DATA

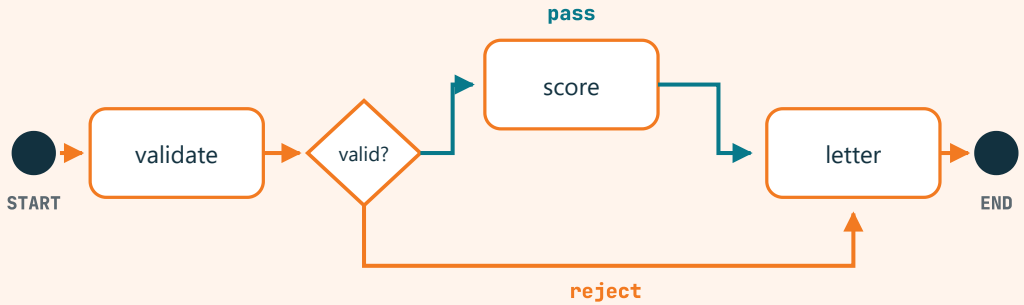


it retrieves the connected subgraph, not the nearest paragraph

NODES	entities: a customer, a loan, a branch, an event
EDGES	relationships that are true in your data
FIXES	retrieval that misses facts two hops away

GRAPH ENGINEERING

a graph of your CODE



it runs functions in an order you drew and can checkpoint

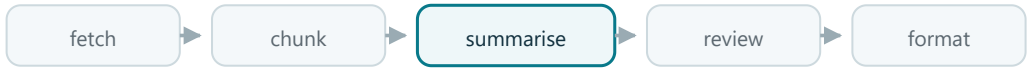
NODES	functions: validate, score, write the letter
EDGES	the order those functions run in
FIXES	work that needs order, parallelism, pause, resume

Most Graphs Have Too Many Nodes

KGP Talkie
LEARN. BUILD. TALK AI

THE MOST COMMON FAILURE

"summarise this PDF", drawn as five nodes



Four of those are one function call, and the fifth is one **model call. You have paid distributed-systems tax for a script.**

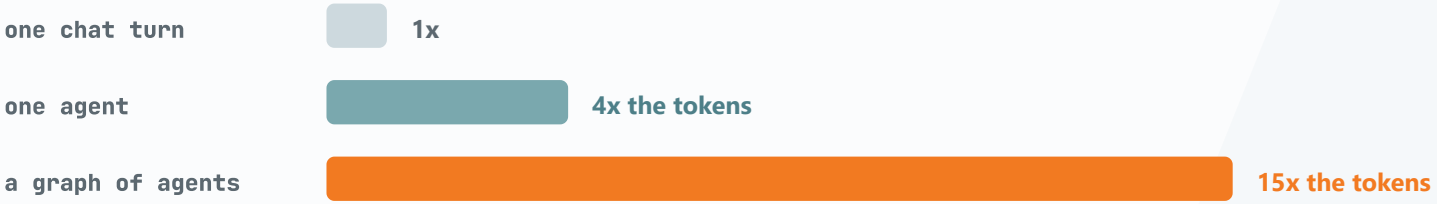
TWO TESTS BEFORE YOU ADD A NODE

The napkin test
If you cannot draw the whole graph on a napkin, it is too complex.

The collapse test
If merging two nodes loses nothing, they were never two nodes.

A node earns its place with a different model, a different toolset, **or a genuinely separate role such as a read-only reviewer.**

AND EVERY NODE YOU ADD MULTIPLIES THE BILL



Anthropic's own numbers. A weak verifier in a graph does not just waste money, it wastes it in parallel.

THE COUNTERWEIGHT

Anthropic's multi-agent research system still beat a single Opus agent by 90.2% on their own eval, because research genuinely fans out.

Twenty Agents That All Agree

LOOKS LIKE peer review, or a second opinion. Same model, same context, same answer.

WHY THE REVIEWER NODE FAILS

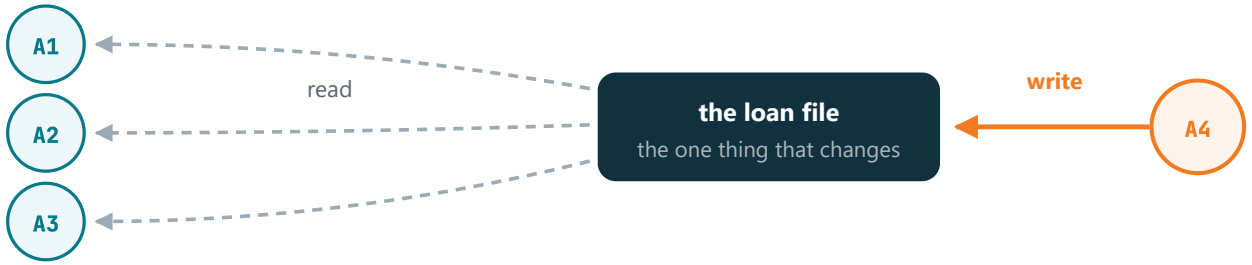


They agree, and they agree confidently, because **models measurably prefer their own output.**

WHAT A REAL REVIEWER NODE NEEDS

1. **A different model**
   not a second call to the one that wrote it
2. **Fresh context**
   the artifact, not the conversation that produced it
3. **Evidence the graph cannot fabricate**
   tests that ran, code that compiled, a balance that moved

THE RULE COGNITION LANDED ON AFTER A YEAR OF RUNNING DEVIN



Reading is safe in parallel, because a bad opinion costs nothing until somebody acts on it.

Writing is where the damage happens, so keep it in one place where you can see it.

Four Things to Keep

KGP Talkie
LEARN. BUILD. TALK AI

Draw it on a napkin

Before any code, and make every node justify itself. If collapsing two nodes loses nothing, they were never two nodes in the first place.

State is the contract

Typed schema, and an explicit decision about which node may write which field. State drift is the main way a graph rots, quietly, from the inside.

Route with code

An edge is a decision, and the question is who makes it. Code routes anything checkable. Spend a model call only where judgement is genuinely needed.

Bring in a stranger

A different model, fresh context, and a verdict anchored to evidence the graph cannot fabricate. Then put a budget cap on every single node.

Is one loop still enough?

YES, SO DO NOT BUILD ONE
Add brakes, a real completion check, and a critic instead.

NO, SO COORDINATION IS THE JOB
Draw it, type the state, and cap every node's budget.